Mathcamp 2026 Qualifying Quiz Problem 5 Solution

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Part a)

First, note that updating a digit m times increases it by 1 modulo k. Assume the starting digit is d. If $d < k-1$, then $d+1 \equiv d \pmod{k}$, and if $d = k-1$, then $0 \equiv k \pmod{k}$. So, starting from d simply makes the digit $d+m$ modulo k.

To reach k^n different strings, we must perform $k^n - 1$ moves. If a digit is increased m times, then the final digit d satisfies $m \equiv d \pmod{k}$. The final digit sum is $n(k-1)$, therefore, we must have

$$k^n - 1 \equiv n(k-1) \pmod{k}$$

$$-1 \equiv -n \pmod{k} \implies n \equiv 1 \pmod{k}.$$

We claim that this condition is sufficient.

Lemma 1: Let $a = a_1a_2a_3\cdots a_m$ be a string of m digits modulo k. For every m, there is a sequence of $k^m - 1$ moves that reaches each m-digit string exactly once, starting at a and ending at $(a_1-1)a_2a_3\cdots a_m$.

Proof: We proceed by induction on m. For the base case $m = 1$, incrementing the single digit $k-1$ times reaches all k digits, ending at $a_1 + k - 1 \equiv a_1 - 1 \pmod{k}$.

For the inductive step, assume Lemma 1 holds for some integer m, and consider an $(m+1)$-digit string $a_1a_2\cdots a_{m+1}$. This means there exists a sequence of moves F_m of length $k^m - 1$ on the positions $1, 2, \ldots, m$, which will reach every m-digit string on those positions, ending at $(a_1-1)a_2\cdots a_m$. Let F'_m denote the same sequence of moves applied to positions $2, 3, \ldots, m+1$. Now consider the sequence

$$F_{m+1} = (F'_m, 1, F'_m, 1, F'_m, \ldots, 1, F'_m),$$

where F'_m is applied a total of k times and between each such sequence we increment the first digit. Initially the first digit is a_1. After a repetitions of F'_m and first-digit increments, the first digit is equal to $a_1 + a \equiv A \pmod{k}$. By the inductive hypothesis, each time we apply F'_m, we cycle through all the length-m suffixes (with first digit fixed) exactly once, so all the strings we reach with the above sequence are distinct. The final string has first digit $a_1 + (k-1) \equiv a_1 - 1 \pmod{k}$ and the other digits are

$$(a_2 - 1 - 1 - \cdots - 1)a_3a_4\cdots a_{m+1} = (a_2 - k)a_3a_4\cdots a_{m+1} = a_2a_3a_4\cdots a_{m+1}$$

modulo k, as required. This completes the induction. $\square$

Now, let S^n be the concatenation of n copies of S, e.g. $(10)^3 = 101010$, and let $K = k^k$.

Lemma 2: For all $m \geq 1$, there is a sequence of $K^m - 1$ moves that reaches each mk-digit string exactly once, starting from 0^{km} and ending at $((k-1)\cdot 0^{k-1})^m$.

Proof: We proceed by induction on m. The base case $m = 1$ is clear from Lemma 1, which allows us to transform from 0^k to $(k-1)\cdot 0^{k-1}$ using a sequence of moves $B = (b_1, b_2, \ldots, b_{K-1})$.

For the inductive step, assume Lemma 2 holds for some integer m with some sequence of moves G_m, and consider the case of a length-$(m+1)k$ string. View the string as $m+1$ blocks of k digits. Let G'_m be the same sequence as G_m, but applied to blocks $2, 3, \ldots, m+1$. Now, consider the sequence

$$G_{m+1} = (G'_m, b_1, G'_m, b_2, G'_m, \ldots, b_{K-1}, G'_m),$$

so that G'_m is applied K times. Since G'_m only affects blocks 2 to $m+1$, each application of G'_m reaches all possible suffixes for the current string in the first block. Therefore, going through the sequence, we will reach all combinations suffixes with block 1 prefixes, so we will reach all length-$(m+1)k$ strings exactly once. The ending string will be

$$((k-1)\cdot 0^{k-1}) \cdot ((k-1)\cdot 0^{k-1})^m = ((k-1)\cdot 0^{k-1})^m,$$

as desired, completing the induction. $\square$

Finally, we construct the sequence of moves for $n \equiv 1 \pmod{k}$. Let $n = mk + 1$ for some integer m. Consider the sequence of moves G_m described in Lemma 2. Notice that by permuting the digit labels by increasing each label by a modulo k, we can display all K^m strings exactly once, but ending at $(0^a(k-1)0^{k-a-1})^m$. Let this family of sequences be $H_{m,a}$. Consider the following sequence of moves:

$$H_{m,1}, n, H_{m,2}, n, H_{m,3}, \ldots, n, H_{m,k}$$

We can see that $H_{m,1}$ lists all K^m strings ending with 0, then incrementing the last digit and applying $H_{m,2}$ lists all K^m strings ending with 1, etc. At the end, the last digit has been incremented $k-1$ times, while the $H_{m,a}$ operations make all digits in the k-blocks equal to $(k-1)$. Therefore, the final string is $(k-1)^n$, as desired.

Part b)

The problem can be rephrased as follows:

There is a directed graph of k^2 vertices, where each vertex has coordinates (x, y) with $0 \leq x, y < k$. Each vertex (x, y) has a directed edge to $(x+1, y)$ and $(x, y+1)$, where coordinates are taken modulo k. Call these edges the vertex's x-edge and y-edge respectively. Starting at $(0, 0)$, we wish to find all possible endings of a Hamiltonian path on this graph.

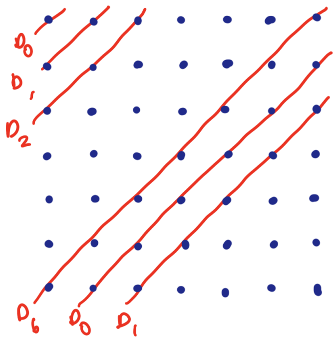
First, notice that each move increases $x + y$ by one modulo k. Since we perform $k^2 - 1$ moves, we see that the ending vertex must satisfy $x + y \equiv k^2 - 1 \equiv k - 1 \pmod{k}$, which are the vertices on the main diagonal opposite $(0, 0)$.

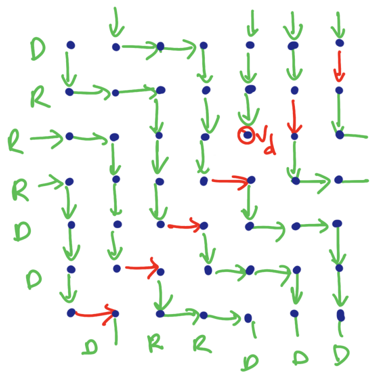
Consider a valid Hamiltonian path on this graph, ending at $v_d = (d, k-1-d)$. Each vertex except for v_d has outwards degree 1, and each vertex except for $(0, 0)$ has inwards degree 1. Let the diagonals be defined as

$$D_t = \{(x, y) \mid x + y \equiv t \pmod{k}\},\ 0 \leq t < k.$$

Notice that v_d lies on diagonal D_{k-1}. First, assume $d > 0$, so $v_d \neq v_0$, and consider the outgoing edge from $v_0 = (0, k-1)$. Since $(0, 0)$ has in-degree 0, v_0 must use the x-edge, so $(1, k-1)$ has in-degree 1. Since the Hamiltonian path cannot enter $(1, k-1)$ again, this forces v_1 to use the x-edge also. We can continue this argument to see that for $0 \leq t < d$, v_t uses the x-edge. Similarly, for $d < t < k$, v_t uses the y-edge.

Also, by the same vertex degree argument, we can see that the vertices in each diagonal either all use the x-edge or all use the y-edge: the edges from D_i to D_{i+1} (where $i < k-1$) must form a bijection between the vertices in the two diagonals.





Let X be the number of diagonals from $D_0, D_1, \ldots, D_{k-2}$ that use x-edges, and let d be defined identically as before (determining the ending vertex). When traversing $D_0, D_1, \ldots, D_{k-2}$, we always increase our x-coordinate by X, so we can construct the following sequence x_i, where x_i tracks our x-coordinate the i-th time we reach a vertex in D_{k-1}:

1. $x_1 = X$, and $0 \leq x_i < k$ for all i.
2. For $i \geq 1$:

   if $x_i < d$, then $x_{i+1} \equiv x_i + 1 + X \pmod{k}$.

   if $x_i > d$, then $x_{i+1} \equiv x_i + X \pmod{k}$.

   if $x_i = d$, the sequence ends as we have reached the target (so x_{i+1} is not defined).

To verify whether v_d is a possible ending vertex for a Hamiltonian path, we wish to determine whether there exists X such that the sequence ends at x_k (thus reaching all vertices). Since all vertices in D_{k-1} have distinct x-coordinates, we see that $x_1, x_2, \ldots, x_k$ must be a cyclic permutation of $\{0, 1, 2, \ldots, k-1\}$. Specifically, we can write it as a composition of two permutations: first, to add one to all elements less than d (and send d to 0 to satisfy being a permutation), and second, one that represents addition of X modulo k. We can use Cauchy's two-line notation to write our permutations, where the initial elements are listed in the top row and their respective images are listed in the bottom row (modulo k):

$$\begin{pmatrix} x_1 & x_2 & \cdots & x_k \\ x_2 & x_3 & \cdots & x_1 \end{pmatrix} = \begin{pmatrix} 0 & 1 & \cdots & d-1 & d & d+1 & \cdots & k-1 \\ 1 & 2 & \cdots & d & 0 & d+1 & \cdots & k-1 \end{pmatrix} \begin{pmatrix} 0 & 1 & \cdots & k-1 \\ X & X+1 & \cdots & X+k-1 \end{pmatrix}$$

Observe the signs of the permutations. Since the left-hand side is a cycle of length k, it has sign $(-1)^{k-1}$. The left permutation on the right-hand side contains a cycle of length $d+1$, so it has sign $(-1)^d$. Finally, the rightmost permutation will form $\gcd(k, X)$ disjoint cycles, each of length $\frac{k}{\gcd(k,X)}$. Therefore, its sign will be

$$\left((-1)^{\frac{k}{\gcd(k,X)}-1}\right)^{\gcd(k,X)} = (-1)^{k-\gcd(k,X)}.$$

Since the sign of a permutation is multiplicative, we have $(-1)^{k-1} = (-1)^{d+k-\gcd(k,X)}$, which gives

$$d - \gcd(k, X) \equiv 1 \pmod 2.$$

If k is even, then $\gcd(k, X)$ can be either parity, so d can have either parity. When k is odd, $\gcd(k, X)$ is always odd, so d must be even.

We now construct a valid Hamiltonian path that ends at all possible shown v_d by selecting suitable X.

Case 1: k is odd.

As found above, we can only reach v_d when d is even. Consider $X = 1$. Following the definition of x_i, the sequence $x_1, x_2, \ldots$ will increase by 2 whenever $x_i < d$, and increase by 1 (modulo k) whenever $x_i > d$, yielding:

$$1, 3, 5, \ldots, d-1, d+1, d+2, d+3, \ldots, k-1, 0, 2, 4, \ldots, d.$$

The sequence forms a permutation of $0, 1, \ldots, k-1$, so setting $X = 1$ produces valid Hamiltonian paths to each ending vertex v_d for even d.

Case 2: k is even.

Subcase 2.1: d is even.

Once again, consider $X = 1$, and follow the steps to produce the sequence x_i:

$$1, 3, 5, \ldots, d-1, d+1, d+2, d+3, \ldots, k-1, 0, 2, 4, \ldots, d.$$

The sequence forms a permutation of $0, 1, \ldots, k-1$, so setting $X = 1$ produces valid Hamiltonian paths to each ending vertex v_d for even d.

Subcase 2.2: d is odd.

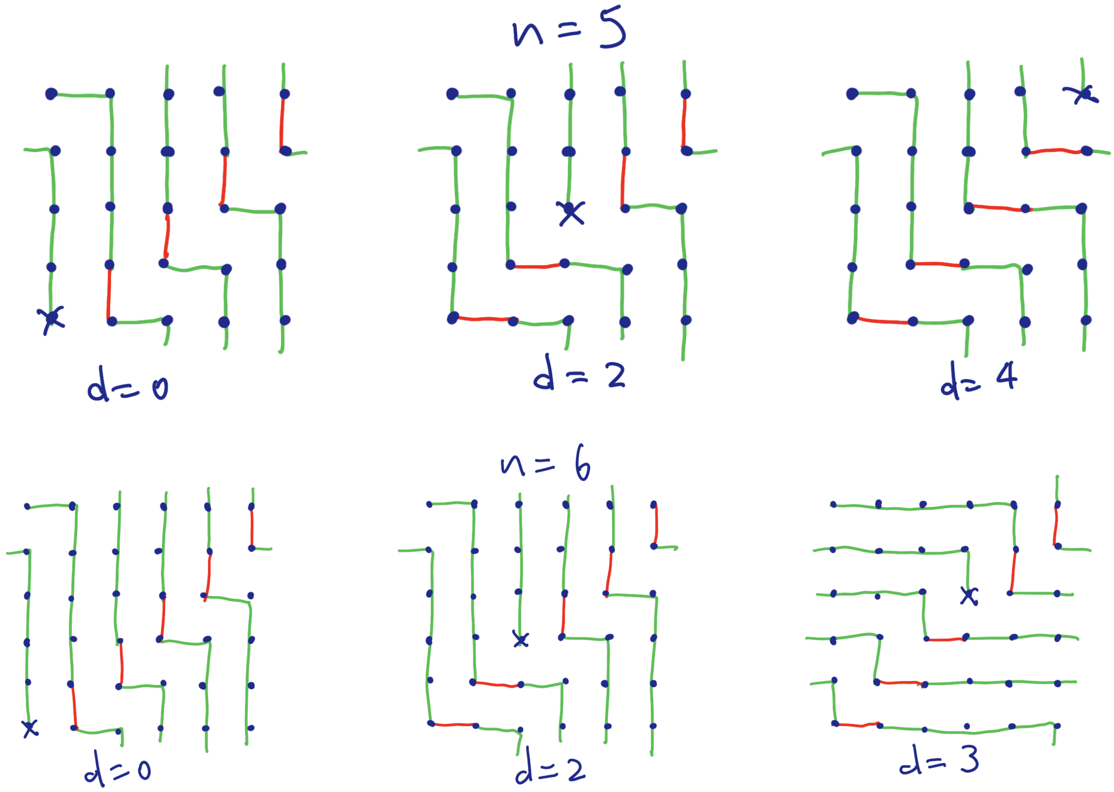
Consider $X = k-2$. Following the definition of x_i, the sequence $x_1, x_2, \ldots$ will decrease by 2 (increase by $k-2 \pmod k$) whenever $x_i > d$, and decrease by 1 modulo k (increase by $k-2+1 \pmod k$) whenever $x_i < d$, yielding:

$$k-2, k-4, k-6, \ldots, d+1, d-1, d-2, d-3, \ldots, 1, 0, k-1, k-3, k-5, \ldots, d.$$

The sequence forms a permutation of $0, 1, \ldots, k-1$, so setting $X = k-2$ produces valid Hamiltonian paths to each ending vertex v_d for odd d.

This finishes the construction. To conclude, the possible ending digits are all pairs of the form $(d, k-1-d)$ for $0 \le d < k$, where d additionally must be even if k is odd.

Some example paths are shown below, using the values of X given above:


n = 5
d = 0
d = 2
d = 4
n = 6
d = 0
d = 2
d = 3